Problem statement 1: Smart Device system.

Classes
- Camera → take Photo()
- Music Player → PlayMusic()
- Smartphone → inherit both.

Output:
- capture Photo
- Playmusic

Problem statement 2: Hybrid Vehicle system.

Classes
- Petrol Engine → petrol info.
- Electric Engine → battery info
- Hybrid car → inherit both.

display:
- fuel type
- Battery level.

Problem statement 3: Student Employee system.

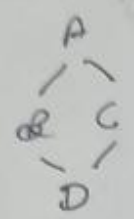
Classes
- student → marks
- Employee → salary
- Intern → inherits both.

display
- marks
- salary
- Internship status.

Real life Example

Teacher → Teaches
Researcher → Researches.
Professor → does both.

Important Problem (Diamond Problem)



If Both B and C inherit from A
Gets two copy of A X
you will learn this later using Vertual Keyword.

Conflicting Example

```
class A {
    public:
        void show() { cout << "A" ; }
    };
class B {
    public:
        void show() { cout << "B" ; }
    };
class C: public A, public B {
    };
int main() {
    C obj;
    // obj.show() X Error ( conflict
```

we can take this

obj.A::show();
obj.B::show()

#Day 32 (Multiple inheritance)

When A class is inherited from more than one parent class

Parent1 + Parent2 → Child

Syntax:

```
class child: public parent1, public parent2 {
  };
```

Example:

```
#include <iostream>
using namespace std;
class A {
  public:
      void showA(){
          cout << "Class A" << endl;
  }
};

class B {
  public:
      void showB(){
          cout << "Class B" << endl;
  }
};

class C: public A, public B{
  };

int main () {
    Cobj;
    obj.showA();
    obj.showB();
  }.
```

Key concept
- Child class gets feature from parent1 and parent2